Optimizing Falcon 9 Mass Consumption and Trajectory

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Prologue

For as long as I can remember, rockets have been a spectacle for me. Space travel, propulsion, and launches captivated my mind and became a passion. After studying the topic out of curiosity and interest, I put together this paper, combining my interests in mathematics, physics, and aerospace. I came to realize that my favorite rocket, the Falcon 9, is not just an intricate system; it is a mathematical and engineering marvel.

1 Formulating the Problem

1.1 Introduction

I modeled the first stage of the Falcon 9 rocket to compare a mathematically optimized trajectory and throttle schedule with the actual flight path it uses. I sourced information from publicly available sources and checked it for feasibility. The goal is to find a trajectory and throttle schedule that minimizes first-stage fuel consumption. This study formulates the optimal-control problem using Pontryagin's Minimum Principle (PMP), derives the corresponding Hamiltonian and necessary optimality conditions, and uses Hermite–Simpson direct collocation to obtain a numerical solution.

1.2 Defining the Environment

The environment is represented as a two-dimensional polar coordinate system, where θ is the downrange angle and r is the rocket's distance from Earth's center. Polar coordinates naturally incorporate Earth's curvature and simplify the trajectory model compared with three-dimensional spherical or cylindrical coordinates. The units of θ are radians and the units of r are meters.

The model uses simplifying assumptions to focus on the trajectory and throttle optimization: a constant specific-heat ratio when calculating nozzle exit pressure; a fixed nozzle exit pressure; instantaneous changes in engine mass-flow rate and pitch rate; exponential atmospheric pressure and density profiles with a scale height of 8500 m; constant engine efficiency; zero angle of attack; pitch rate as the pitch control; a perfectly spherical Earth; and idealized physics and chemistry. The thrust model retains the pressure-thrust contribution, $(P_e - P_a(r))A_n$, while treating P_e as constant and ambient pressure $P_a(r)$ as altitude-dependent. Thus, the model does not impose perfect expansion at every altitude during ascent.

1.3 Variables

The five state variables are:

- $r(t)$: distance from Earth's center;
- $\theta(t)$: downrange angle from the initial position;
- $v(t)$: rocket speed;
- $\gamma(t)$: pitch angle measured relative to the local horizon;
- $m(t)$: rocket mass.

The two control variables are $\omega(t) = \dot{\gamma}(t)$, the pitch rate, and $\dot{m}_f(t)$, the positive fuel mass-flow rate.

1.4 Initial and Terminal Conditions

The initial conditions are

$$r(0) = R_E, \qquad R_E = 6.378 \times 10^6 \text{ m}, \tag{1}$$

$$v(0) = 0, \qquad \gamma(0) = \frac{\pi}{2}, \tag{2}$$

$$\theta(0) = 0, \qquad m(0) = 549054 \text{ kg}. \tag{3}$$

The terminal conditions are

$$r(t_f) = R_E + 70000 \text{ m}, \tag{4}$$

$$v(t_f) \geq 2000 \text{ m/s}, \tag{5}$$

$$\frac{\pi}{10} \leq \gamma(t_f) \leq \frac{\pi}{4}. \tag{6}$$

The target altitude of 70 km approximates a representative main-engine cutoff (MECO) altitude, although actual stage separations occur at varying altitudes. The minimum terminal speed of 2000 m/s provides a target for continued ascent with a second stage. The terminal pitch interval allows some freedom while keeping the final orientation within the selected range.

The control bounds are

$$1566 \leq \dot{m}_f(t) \leq 2736 \quad \text{kg/s}, \tag{7}$$

$$-0.0175 \leq \omega(t) \leq 0.0087 \quad \text{rad/s}. \tag{8}$$

The optimization problem is to maximize final mass, equivalently minimizing total fuel consumed for fixed initial mass:

$$\max_{\dot{m}_f(\cdot), \omega(\cdot), t_f} m(t_f), \tag{9}$$

subject to the terminal conditions, control bounds, and state dynamics above.

1.5 General Formulas and Constants

The thrust model is

$$T = \dot{m}_f v_e + (P_e - P_a(r))A_n. \tag{10}$$

Gravity magnitude, drag, ambient pressure, and density are modeled as

$$g(r) = \frac{\mu}{r^2}, \tag{11}$$

$$D(r, v) = \frac{1}{2}\rho(r)v^2 A_c C_d, \tag{12}$$

$$P_a(r) = 101325 \exp\left(-\frac{r - R_E}{8500}\right), \tag{13}$$

$$\rho(r) = 1.225 \exp\left(-\frac{r - R_E}{8500}\right). \tag{14}$$

The constants used are summarized below.

Parameter	Value
Earth radius, R_E	6.378×10^6 m
Effective exhaust velocity, v_e	2776 m/s
Nozzle exit pressure, P_e	73168 Pa
Nozzle exit area, A_n	6.786 m^2
Earth gravitational parameter, μ	3.986×10^{14} m^3/s^2
Rocket cross-sectional area, A_c	3.423π m^2
Drag coefficient, C_d	0.32 (estimated)

1.6 State Equations

The state dynamics are

$$\dot{r} = v \sin\gamma, \tag{15}$$

$$\dot{\theta} = \frac{v \cos\gamma}{r}, \tag{16}$$

$$\dot{v} = \frac{T - D}{m} - \frac{\mu}{r^2}\sin\gamma, \tag{17}$$

$$\dot{\gamma} = \omega, \tag{18}$$

$$\dot{m} = -\dot{m}_f. \tag{19}$$

The radial rate is the vertical component of velocity, and the angular rate follows from the tangential component, $v\cos\gamma = r\dot{\theta}$. The speed equation accounts for thrust, drag, and the gravity component opposing the radial component of motion. The pitch state follows the prescribed pitch-rate control, and vehicle mass decreases at the fuel mass-flow rate.

2 Solving Using Pontryagin's Minimum Principle

2.1 Pontryagin's Minimum Principle

PMP gives necessary conditions that an optimal control must satisfy. These include the state equations, costate equations, a Hamiltonian optimality condition, and boundary conditions. In general, for $\dot{x} = f(x, u, t)$ and a running cost $L(x, u, t)$, the Hamiltonian is

$$H(x, u, \lambda, t) = L(x, u, t) + \lambda^T f(x, u, t), \tag{20}$$

where λ is the vector of costates. The costates satisfy $\dot{\lambda} = -\partial H/\partial x$, and the optimal control minimizes the Hamiltonian over the admissible control set.

For this problem, set $u_1 = \omega$ and $u_2 = \dot{m}_f$. Since

$$m(t_f) = m(0) - \int_0^{t_f} \dot{m}_f(t)\, dt, \tag{21}$$

maximizing final mass is equivalent to minimizing

$$J = \int_0^{t_f} u_2(t)\, dt. \tag{22}$$

The Hamiltonian is

$$H = u_2 + \lambda_r \dot{r} + \lambda_\theta \dot{\theta} + \lambda_v \dot{v} + \lambda_\gamma \dot{\gamma} + \lambda_m \dot{m}. \tag{23}$$

Substituting the state equations gives

$$\begin{aligned} H = {} & u_2 + \lambda_r v \sin\gamma + \lambda_\theta \frac{v\cos\gamma}{r} \\ & + \lambda_v \left[\frac{u_2 v_e + (P_e - P_a(r))A_n - D(r, v)}{m} - \frac{\mu}{r^2}\sin\gamma \right] \\ & + \lambda_\gamma u_1 - \lambda_m u_2. \end{aligned} \tag{24}$$

2.2 Minimizing the Hamiltonian

The optimal control is selected by minimizing H over the admissible control set. Because the Hamiltonian is linear in the controls under this formulation, the switching functions determine whether a control lies at its lower or upper bound. For the mass-flow control, the switching function is

$$\Phi_{u_2} = \frac{\partial H}{\partial u_2} = 1 + \lambda_v \frac{v_e}{m} - \lambda_m. \tag{25}$$

For pitch rate,

$$\Phi_{u_1} = \frac{\partial H}{\partial u_1} = \lambda_\gamma. \tag{26}$$

For a nonzero mass-flow switching function, the minimizing bang-bang control is

$$u_2^* = \begin{cases} u_{2,\min}, & \Phi_{u_2} > 0, \\ u_{2,\max}, & \Phi_{u_2} < 0. \end{cases} \tag{27}$$

The corresponding switching times occur where the switching function changes sign. The costates are needed to determine those times.

2.3 Costate Equations

The costates follow $\dot{\lambda}_i = -\partial H/\partial x_i$. In expanded form, using the model above,

$$\dot{\lambda}_r = -\frac{\partial H}{\partial r}, \qquad \dot{\lambda}_\theta = -\frac{\partial H}{\partial \theta} = 0, \tag{28}$$

$$\dot{\lambda}_v = -\frac{\partial H}{\partial v}, \qquad \dot{\lambda}_\gamma = -\frac{\partial H}{\partial \gamma}, \tag{29}$$

$$\dot{\lambda}_m = -\frac{\partial H}{\partial m}. \tag{30}$$

The derivatives include the radial dependence of atmospheric pressure and density, the inverse-square gravity term, and the dependence of acceleration on mass. The resulting nonlinear boundary-value problem is difficult to solve directly, so the numerical solution is obtained using direct collocation.

2.4 Limitations of PMP

PMP is useful for deriving necessary conditions, but solving the coupled nonlinear state and costate differential equations with the boundary conditions and control bounds can be impractical. A numerical method, direct collocation, is therefore used to obtain an approximate solution.

3 Solving Using Direct Collocation

3.1 Direct Collocation

Direct collocation discretizes the trajectory into a finite set of time points and optimizes state and control values at those points. Constraints enforce consistency with the state equations. In the Hermite–Simpson method, the midpoint state is approximated from the interval endpoints, and the difference between the integrated dynamics and the state change is constrained to zero through a defect equation. The resulting nonlinear program is passed to IPOPT (Interior Point OPTimizer).

3.2 Collocation Inputs

The solver uses the initial and terminal conditions and control bounds specified above, together with the dynamics in Eqs. (15)–(19). Additional physical constraints are $r \geq R_E$, $v \geq 0$, and $m > 0$, with final time constrained by $100 \leq t_f \leq 500$ s to avoid unrealistic solutions. The numerical code exports state and control values for analysis; the data table and five-launch MECO comparison data are linked in Appendix E.

3.3 Results of the Collocation

The solver converged to a trajectory with a final time of approximately 140 s, terminal altitude of 70 km, and terminal speed of 2000 m/s. The mass-flow control switches at approximately 124 s. The figures below reproduce the solver plots supplied with the draft.

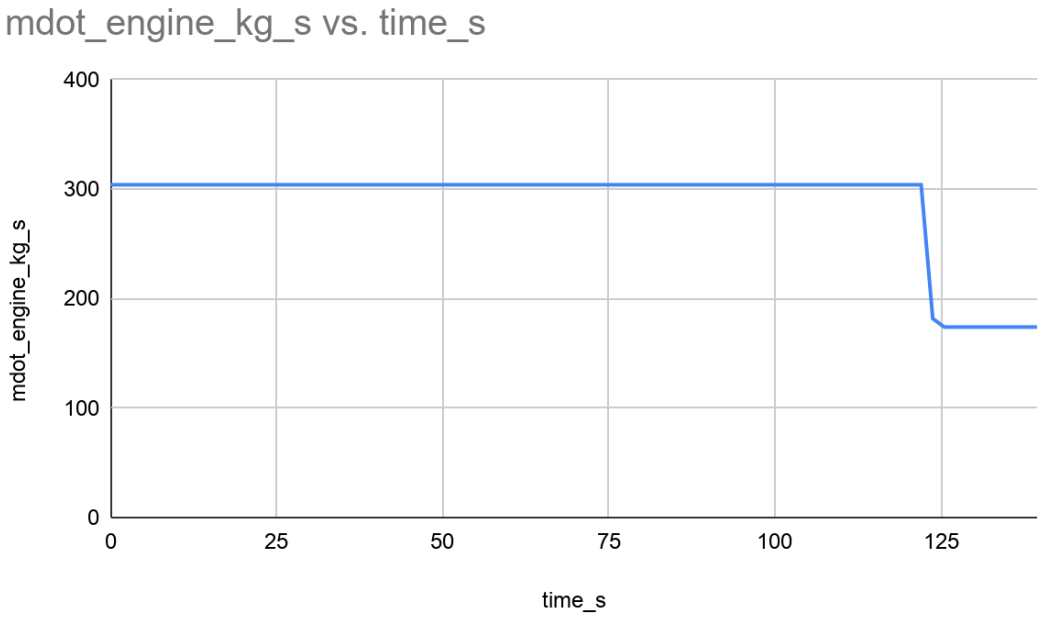


Figure 1: Fuel mass-flow rate versus time.

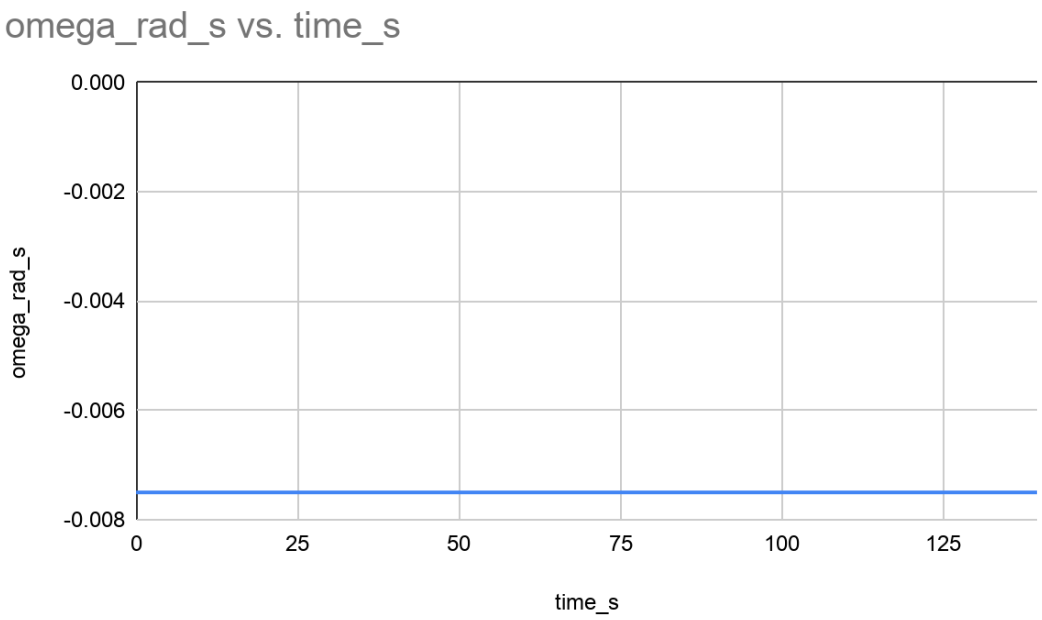


Figure 2: Pitch-rate control versus time.

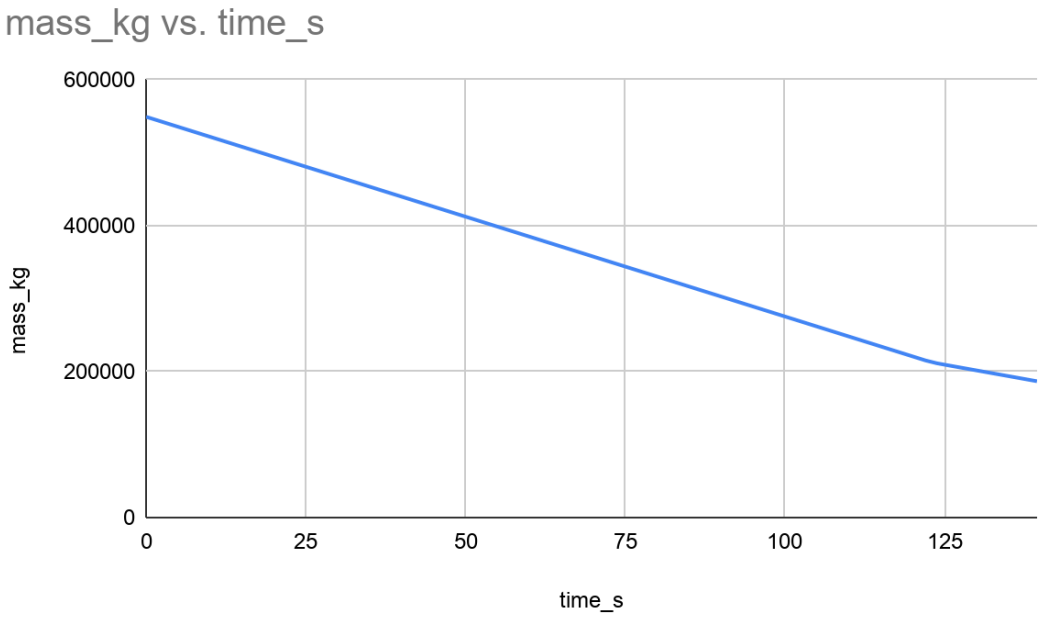


Figure 3: Rocket mass versus time.

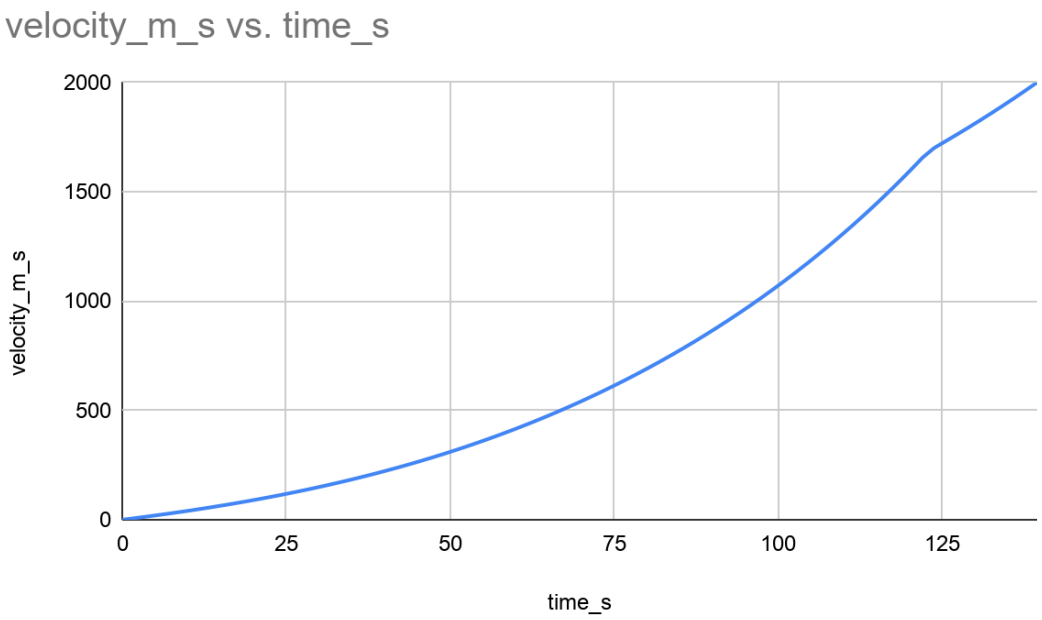


Figure 4: Rocket speed versus time.

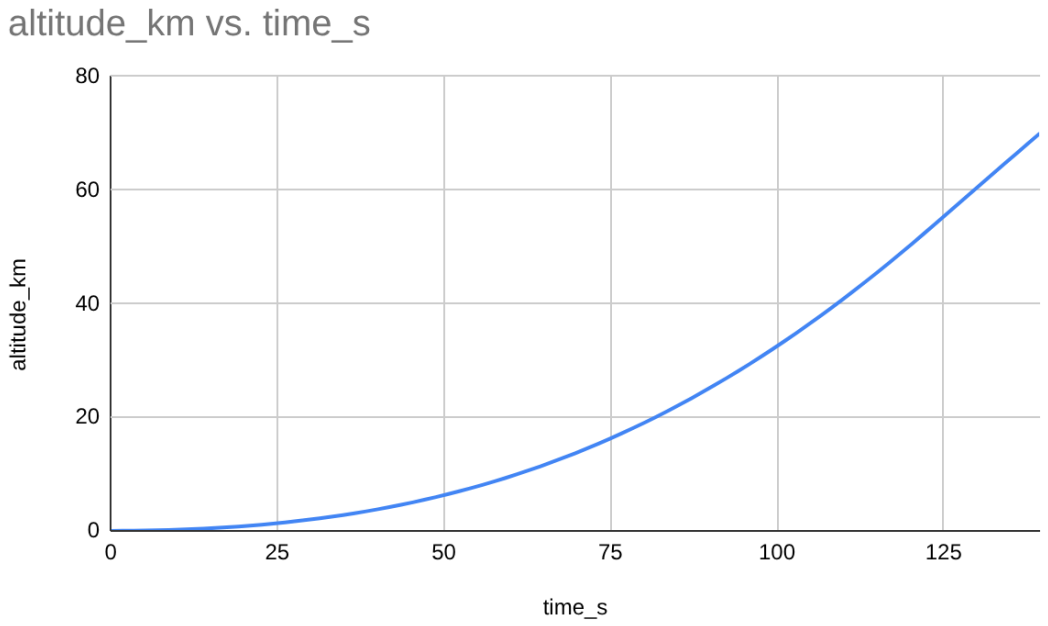


Figure 5: Altitude versus time.

Based on the mass-flow plot, the control exhibits bang-bang behavior, switching at approximately $t = 124$ s. The pitch-rate plot shows a nearly constant value over the interval. The terminal altitude and speed meet the specified terminal constraints. The model reports 362 320 kg of fuel consumed by MECO, corresponding to approximately 88.2–91.5% of the two cited estimates of total first-stage propellant. The optimized flight time is about 2 minutes and 20 seconds.

3.4 Comparison with Falcon 9 Flight Data

Five livestreamed Falcon 9 launches were analyzed and averaged to estimate MECO time, speed, and altitude. The average values were approximately 149 s, 2124 m/s, and 67 km, respectively. The optimized model reaches MECO about 9 seconds earlier than this average, with a terminal speed within 6% and altitude within 4.5% of the averages. These comparisons show that the solution satisfies the model's selected conditions and is in the same general range as the observed flight data; they do not by themselves establish complete physical feasibility.

The cited propellant mass of the Falcon 9 first stage is either 395 700 kg or 410 900 kg, depending on the source. Using the lower value, the model leaves approximately 33 380 kg, or 8.43%, of the initial propellant estimate at MECO. Using the higher value leaves approximately 48 580 kg, or 11.82%. Bjarnø (2015) estimated that a downrange landing would require approximately 20 000 kg, or about 6% of the propellant reserves. Barker et al. (2024) estimated that re-entry and landing require slightly less than 7% of the fuel reserves. These figures are approximate comparisons, not a verified demonstration that the optimized trajectory has enough propellant to complete recovery.

One possible reason the remaining propellant differs from observed operations is that the model uses simplified dynamics and assumptions rather than a complete flight and recovery simulation. The Falcon 9 has also undergone repeated design and operational improvements over many flights.

The present model is intended to explore how trajectory and throttle controls affect propellant use, not to provide a flight-ready control schedule.

4 Conclusion

This study formulated a Falcon 9 first-stage optimal-control problem using Pontryagin's Minimum Principle and solved a discretized version using Hermite–Simpson direct collocation. The numerical solution reached the selected MECO altitude and minimum speed while optimizing fuel consumption through the throttle schedule and pitch-rate control. The resulting time, speed, and altitude are broadly comparable with the averages obtained from five analyzed launches. The propellant remaining at MECO is estimated at 8.43–11.82% depending on which published first-stage propellant mass is used. This remaining amount is only an estimate and does not independently prove downrange landing feasibility.

4.1 Possible Model Setbacks

Simplifying assumptions and omitted physical effects limit the model's accuracy. A mathematically feasible solution under the chosen equations and constraints is not necessarily physically feasible in the real world. The model is best understood as an exploration of the relationship between throttle schedule, trajectory, and propellant consumption rather than as a practical control law for a real rocket.

4.2 Improvements

Future work could simulate the optimized controls in a higher-fidelity physics environment and test whether they achieve the intended terminal conditions. A two-point boundary-value-problem solver could also be used to solve the nonlinear costate equations and identify switching functions more directly, although convergence may be difficult.

References

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A Terminal Pitch Bounds

The terminal pitch bounds are $\pi/10 \le \gamma(t_f) \le \pi/4$. The upper bound is associated with a substantial horizontal velocity component, while the lower bound preserves some vertical velocity component and leaves the final pitch relatively unconstrained.

B Mass-Flow Bounds

According to the cited Falcon 9 Full Thrust source, the throttle capability ranges from approximately 482 kN to 845 kN per engine. The ratio is $482/845 \approx 57.0\%$. A cited Merlin engine source also reports a 57% throttle capability. With a maximum mass-flow estimate of 304 kg/s per engine, the minimum is estimated as $304(0.57) = 173.28$ kg/s. The model uses 174 kg/s per engine to provide a small margin. For nine engines, the bounds become $9(174) = 1566$ kg/s and $9(304) = 2736$ kg/s.

C Pitch-Rate Bounds

The pitch-rate bounds are estimates rather than values derived from Falcon 9 flight data. The lower bound, -0.0175 rad/s, corresponds to approximately -1 degree per second. The upper bound, 0.0087 rad/s, corresponds to approximately 0.5 degrees per second. These bounds limit the rate at which the pitch angle can change.

D Exhaust Velocity and Exit Pressure

The specific impulse relation is

$$I_{sp} = \frac{v_{eq}}{g_0}, \qquad v_{eq} = g_0 I_{sp}. \tag{31}$$

Using $I_{sp} \approx 283$ s and $g_0 = 9.81\,\mathrm{m/s^2}$ gives $v_{eq} \approx 2776$ m/s. In the thrust equation, this value is used for the exhaust-velocity term. The model also retains a pressure-thrust term with constant exit pressure P_e and altitude-dependent ambient pressure $P_a(r)$; these are distinct modeling choices.

To estimate exit pressure, the draft uses the isentropic nozzle relation

$$P_e = P_c \left(1 + \frac{\delta - 1}{2} M_e^2\right)^{-\delta/(\delta-1)}, \tag{32}$$

where $P_c = 10.8$ MPa, M_e is exit Mach number, and δ is the specific-heat ratio. Using an expansion ratio of 16 and the estimate $\delta = 1.20$ gives $M_e \approx 3.604$ and $P_e \approx 73\,168$ Pa. The value of δ is an estimate and should be supported with an appropriate technical source if the paper is submitted for formal review.

E Solver and MECO Data

The direct-collocation data table and the five-launch MECO data are available at:
https://github.com/Yilmates/Optimizing-Falcon-9-Mass-Consumption-and-Trajectory/blob/main/falcon9_optimal_trajectory-139.xlsx

F Remaining Propellant Calculation

The cited sources give two estimates for the first-stage propellant mass: 395 700 kg and 410 900 kg. The model reports 362 320 kg consumed by MECO.

$$\text{Case 1: } \frac{395700 - 362320}{395700} = 0.0843 \approx 8.43\% \text{ remaining},$$
$$1 - 0.0843 = 91.57\% \text{ consumed};$$

$$\text{Case 2: } \frac{410900 - 362320}{410900} = 0.1182 \approx 11.82\% \text{ remaining},$$
$$1 - 0.1182 = 88.18\% \text{ consumed}.$$